

PILOT STUDY ON THE EFFECTIVENESS OF SYSTEMATIC DESENSITIZATION WITH NEUROLOGICALLY IMPAIRED CHILDREN WITH PHOBIC DISORDERS

MARTIN OBLER AND ROBERT F. TERWILLIGER¹

New School for Social Research

A modified version of Wolpe's systematic desensitization therapy involving direct confrontation with the fear-inducing stimulus was attempted with neurologically impaired children with phobic symptoms. Two hypotheses were tested: (a) A nonverbal therapeutic technique not requiring motivation will produce successful symptom reduction for these children, and (b) awareness of therapeutic procedure is not necessary for successful results. Both hypotheses were confirmed. The implications of these results for further research on systematic desensitization therapy were discussed.

Certain traditional psychotherapies have operated on the theoretical assumption that patient-therapist interaction is a necessary prerequisite in the modification of undesirable behavior (Wolberg, 1954). Elaborate verbal exchange is assumed to be the mechanism through which modification occurs. The patient-therapist exchange focuses on behavior conflicts confronting the patients from present and past experiences. The role of the therapist is crucial in accomplishing a modification of the conflicting behavior. He uses the patient's free associations, transference projections, and countertransference responses as his tools (Freud, 1965). Without a therapist as an interpreter of the conflicting experiences, modification of the undesirable behavior would not occur.

In recent years, a large number of clinical investigations have deviated from the assumptions of the above model. One method of treatment employed by Wolpe (1958), systematic desensitization, has been effective in reducing undesirable responses in adults (Rachman, 1967). Recent clinical evidence has supported its successful use with monophobia (Wolpe, 1962a, 1962b), delinquency (Zimmerman & Zimmerman, 1962), enuresis (Lang & Melamed, 1969; Werry, 1966), maladaptive rage and anxiety (Bandura, 1961), etc. Moreover, Wolpe (1962a, 1962b) describes a treatment procedure in which a medical student was substituted for the on-

going therapist with no disruption occurring in the treatment of a phobic disorder. No apparent patient resistance occurred because of the transfer in therapists. Lazarus and Rachman (1957) report data indicating that a change of therapist enhanced treatment success by reducing the negative associations to a particular therapist.

Systematic desensitization therapy, based on Wolpe's reciprocal inhibition theory, requires little verbal capacity or treatment motivation.² A short examination of the theoretical assumptions demonstrates the logic of the above position. The theory assumes that maladaptive, fearful responses may be eliminated by the presentation of some other response presumably antagonistic to the fearful one. Whether the therapist uses muscle relaxation, classical conditioning of a neutral response, or operant reinforcement as a presumed fear antagonist, the "counterbalancing" principle of the therapy remains the same. By simply developing a counteracting response to the original fear-producing stimulus, we condition a reduction in the fear response. A patient may cognitively perceive the connection between the counteracting response and the fear-producing stimulus, but it is not a necessary requirement. With adults, Wolpe uses muscle relaxation as the conditioned response, but any reinforcement producing the same results could just as easily be substituted. In-

¹Requests for reprints should be sent to Robert Terwilliger, Graduate Faculty of Political and Social Sciences, New School for Social Research, 66 West 12th Street, New York, New York 10011.

²Wolpe has extensively developed this theory in numerous studies he has conducted. The most elaborate description of the theory remains his original work, *Psychotherapy by Reciprocal Inhibition* (1958).

stead of developing a muscle relaxation schedule, the therapist and patient might develop a touch reminder schedule. Each time the patient feels the onset of fear, he automatically responds with the conditioned response of touching himself to remind him of the inappropriateness of this reaction. When the original fear-producing stimulus no longer elicits a fear response, the therapy moves to a new stimulus previously established on a hierarchy of low-intensity to high-intensity fear responses. Relaxation or reminder responses are thus hypothesized to become effective in reducing the strength of the unconditioned stimulus. Furthermore, response generalization is hypothesized to occur and effectively operate in the patient's real-life situations (Wolpe, 1958).

The application of traditional therapeutic techniques to the treatment of childhood disorders has had to face the problem of the relatively low verbal ability of children (Freud, 1965). This has been met by various play therapy techniques among others (Subotnick, 1966). However, these more traditional techniques depend on attacking the "causes" of the disorder for their effectiveness. This approach assumes that somehow, the child develops insight or awareness into the nature of his problem. Furthermore, this insight requires some form of transference or intense interpersonal relationship with the therapist (Subotnick, 1966). If the disturbance becomes severe enough—as in the case of neurological impairment—it may become increasingly difficult to arrive at this requisite state of awareness due to the patient's loss of verbal and abstract ability. Thus, as Eysenck (1952) and Rachman (1965) have suggested, there would be limits to the effectiveness of traditionally oriented therapies in the case of severe emotional or neurological disturbances.

The behavior therapy techniques, rather than attacking the presumed causes of the disturbance, attack the symptoms themselves. These techniques make no assumptions about the necessity of awareness of a transference relationship. This factor alone would suggest the desirability of applying these techniques to the severely disturbed child. While various attempts have been made to apply behavior therapy techniques to children, there have

been very few attempts to use the specific technique of systematic desensitization described above (Ferster & Simons, 1966; Lovaas, Freitag, Gold, & Kassarla, 1965). However, the literature which exists does suggest that systematic desensitization could be an effective technique in the treatment of phobias in children (Jensen & Wamach, 1967; Lazarus, 1958). Rachman (1965) further points out that its use with severe chronic and neurologically impaired persons is limited but promising. He suggests broadening the use of systematic desensitization to these populations.

This study was conducted to further test the validity of applying systematic desensitization to the treatment of severely disturbed children, specifically to a population hitherto unstudied—neurologically impaired children. It would appear that systematic desensitization, which attempts to attack symptoms directly without requiring any great verbal skill or abstract ability on the part of the patient, would be very well suited to the treatment of a population of neurologically impaired children.

METHOD

Subject Selection

Thirty emotionally disturbed, neurologically impaired children were selected to participate in the study. They were drawn from a population of 150 children diagnosed as having minimal brain dysfunction with severe monophobic disorder. The monophobic disorder was one of two types: either (a) phobia regarding the use of a public bus or (b) phobia regarding the sight of a live dog. The Ss having both phobias were excluded from the study. The diagnoses were supplied by the clinic currently servicing the children medically.³ Fifteen Ss were selected at random for the treatment condition (E group) and exposed to a modified version of the Wolpe desensitization method developed by the authors. The remaining 15 Ss served as controls (C group) and were matched with E group Ss for age, sex, intelligence, and phobic diagnosis. Each E group S was matched with an individual therapist by placing the name of all therapists in a bowl with corresponding numbers for each S. A number and name were selected simultaneously, with the

³ The authors wish to thank the New York Association for Brain Injured Children, Brooklyn Chapter, for making available the Ss for the research as well as the appropriate diagnostic material. In addition, partial funding for the pilot study was given through their auspices.

chosen pair remaining together throughout the 10-week period of treatment.

Therapist Training

All 15 therapists were college graduates having no previous training or experience with neurologically impaired children or psychotherapeutic techniques. They were trained by the authors to apply a modified version of the Wolpe method and were instructed not to deviate from the training procedures throughout the course of treatment. The training consisted of two five-hour sessions during a two-week period, in which they were intensively instructed in how to dispense the rewards in attempting to decondition the monophobia. Each therapist was remunerated at the rate of \$5 per five-hour session.

Design

Therapist and *S* met for the first time at the initial therapy session. Each session consisted of a five-hour weekly meeting over a 10-week period. Treatment consisted of exposing each *S* to the fear-inducing stimulus as it appeared in his everyday life. Typically, in desensitization therapy, the patient is asked to imagine the stimulus without actually being exposed to it. The reinforcement then occurs by teaching him to relax in the face of the imagined stimulus. With the relatively nonverbal, unimagined *Ss*, this procedure was out of the question. The modification developed by the authors placed *S* in direct confrontation with the fear-producing stimulus, but with the therapist acting as a buffer between the two. That is, the therapist protected *S* from intense anxiety emerging from exposure to the real phobic stimulus, and rewarded him by approval for successful tolerance of it.

Prior to exposure to the real phobic stimulus, *Ss* were asked to look at pictures or models of the fear-inducing stimulus (bus or dog). This eliminated the need for *S* to imagine the stimulus. Cooke (1966) showed that actual exposure to the fear-inducing stimulus was more effective than merely imagining the stimulus. When *S*'s behavior indicated a tolerance of the picture or model, he was then exposed to the actual fear-producing stimulus (bus or dog). The therapist continuously rewarded *S* through encouragement for moving closer and closer to the object. Eventually, *Ss* were able to move up the scale to more anxiety-producing stimulus (e.g., touching the bus or dog). At this point, *S* was given the option of exposure to the stimulus without the presence of the therapist. A new reward was offered at this time which included toys, books, pets, and candy chosen by *S* at an earlier session with the therapist.

These rewards were dispensed immediately at the time of successful completion of the defined task (e.g., talking to the bus driver, putting a token in the box, staying in the room with a dog). If an *S* overcame the phobia prior to completion of the tenth session, he continued to be reinforced by the therapist for his success. Successful handling of the fear-inducing stimulus was considered to be its generalization to *S*'s daily life without the presence of the therapist. This was measured by a parent report

scale administered to all *E* and *C* group members prior to and at the completion of the treatment period.

It should be clear that the technique used in this study differs from the traditional systematic desensitization technique in four ways: (a) There is no training in relaxation involved in this study.⁴ (b) There was no attempt to ask *S* to develop a hierarchy of anxiety-producing situations to then be treated from the least to the most severe. In effect, a hierarchy was used, but it was determined arbitrarily by the *E*. (c) It was assumed that reinforcement per se was sufficient to bring about removal of the phobic symptom. No assumption about a link between the autonomic nervous system and relaxation was required or made. (d) There was no requirement or demand made of the *S* to indicate verbally the experience of lowered anxiety. Only overt behavioral manifestations were used, with the therapist making the judgment as to whether or not anxiety had been lowered and tolerance of the fear-producing stimulus shown.

Measurement of Treatment Success—Aware and Unaware Subjects

As indicated earlier, the authors' contention was that intellectual awareness of treatment procedure is not a necessity for insuring success in treatment. Measurement of treatment awareness, it was assumed, might be reflected in *S*'s IQ score (based on the Wechsler Intelligence Scale for Children). An arbitrary cutoff point (a score of 75) was established as the critical value. Those *Ss* with scores below a 75 IQ were assumed to be minimally aware of treatment procedures, while those above 75 were considered more likely to be aware. The assumption that IQ is highly correlated with awareness was based on the observable behavior of those *Ss* with an IQ score below 75. These *Ss* appeared oblivious to most stimulation in their immediate environment, and were extremely detached and relatively indifferent to the influence of others on them. The intelligence test was administered to each *S* prior to selection for placement in the *E* and *C* groups. Ten *E* and nine *C* *Ss* scored above 75. There were, therefore, five minimally aware *E* *Ss* and six minimally aware *C* *Ss* (Table 1). The authors recognize the arbitrary nature of selecting an IQ score as a measure of treatment awareness. However, it appears reasonable that a child with an IQ score below 75 functions on a level of very limited awareness and could therefore be used as a standard for a minimum likelihood of awareness of treatment procedures.⁵

The basic measure for treatment effectiveness was a parent rating scale administered to each *S*'s parents prior to and at the completion of treatment. The

⁴ This is contrary to Lamont and Edwards' (1967) finding that relaxation training is necessary for successfully reducing phobias.

⁵ The authors are currently developing a more refined technique for determining an *S*'s awareness of treatment.

scale consisted of 10 questions concerning the child's functioning in society, 1 of which tapped the critical phobic behavior. This question was as follows:

Bus Phobia:

- A. Has your child recently been able to ride a bus? 0—No. 1—Yes, when accompanied by another person. 2—Yes, completely on his own.

Dog Phobia:

- B. Has your child recently been able to touch a dog? 0—No. 1—Yes, with the help of another person. 2—Yes, completely on his own.

The remaining questions dealt with S's functioning in school, with peers, etc., and were included to prevent a favorable response by parents desiring their child's continued participation in the program. The parents were not aware of the purpose of the research and assumed that their child attended a

TABLE 3
PROPORTION OF SUCCESS OF AT LEAST "2"

IQ group	Treated	Untreated
Below 75	3/5	0/6
Above 75	5/10	0/9
Total	8/15	0/15

weekly recreation program. The parent rating was used as the best measure of treatment success on the belief that a person close to a phobic patient is in a good position to observe any reduction of symptoms that generalize to his daily life. All Ss received a 0 rating on an administration of the rating scale made prior to treatment.

RESULTS

The after-treatment measure for both groups appears in Table 1. A score of 1 indicates a reduction in the phobic system sufficient to insure S's ability (as reported by the parents) to ride a bus or touch a dog with the help of another person. A score of 2 indicates this ability without any need for assistance. Each S in the treated condition (E group) received at least a "1" on his parent's rating and in a great many cases a "2" rating. The corresponding C group, on the other hand, showed a disproportionately small increase in "1" ratings and no "2" ratings at all. Table 2 shows that the proportion of successes of at least "1" for treated Ss was significantly higher than for untreated Ss. The critical ratio value for the difference in proportions was 4.4 which is significant at the .00003 level. Table 3 indicates the proportion of successes of at least "2." The critical ratio value for the difference in proportions was 3.3 which is significant at the .0005 level, again an indication of the superiority of the treatment condition in reducing phobic disturbances.

For behavior changes rated as "1," Table 2 indicated 100% success for both high- and low-IQ treatment groups, thereby indicating no difference in treatment success due to conditions of "awareness." For rating "2," Table 3 yields a critical ratio of .04, further demonstrating no difference between "awareness" conditions.

DISCUSSION

The results indicate that treated Ss were significantly affected by the modified system-

TABLE 1

COMPARISON OF IQ SCORES, AGE, AND PARENT RATING FOR TREATED AND UNTREATED SUBJECTS

Subjects	Treated			Untreated		
	Age	IQ	Score at completion of treatment	Age	IQ	Score at completion of treatment
Unaware						
1	7-1	66	2	7-3	67	0
2	9-5	65	1	9-3	68	0
3	9-6	71	1	9-8	64	0
4	10-2	50	2	10-7	73	1
5	10-5	68	2	10-5	57	0
6				9-3	60	0
Aware						
1	10-1	85	1	10-0	86	0
2	12-0	80	2	11-9	81	0
3	9-3	115	2	9-3	146	0
4	7-6	107	2	7-8	106	0
5	9-4	96	2	9-6	94	1
6	10-3	79	1	10-6	81	0
7	11-1	102	1	10-7	103	1
8	9-2	95	1	9-4	96	0
9	10-3	97	2	10-5	95	0
10	11-4	99	1			

TABLE 2

PROPORTION OF SUCCESS OF AT LEAST "1"

IQ group	Treated	Untreated
Below 75	5/5	1/6
Above 75	10/10	2/9
Total	15/15	3/15

atic desensitization method. Control group Ss, on the other hand, showed no significant change in reduction of their phobic symptoms. The S's intellectual ability, presumably indicating the possibility of his awareness of being treated, did not affect the results. This lack of difference is interesting considering the fact that severely neurologically impaired children often receive no treatment because of their presumed lack of motivation and concomitant unawareness (Johnson & Myklebust, 1969). Systematic desensitization in its modified form appears to overcome this difficulty, at least for children suffering from specific dog and bus phobias.

The method of treatment employed in this study was designed for specific phobic disorders. However, its application to other childhood disorders would be a relatively simple task. The identification of the fear-inducing stimulus would be followed by the development of a hierarchy of conditions under which the stimulus appears in the patient's life. Fear-antagonist responses would then be introduced conjunctively with the appropriate reinforcement schedule. Recent studies have demonstrated the effective use of systematic desensitization on such disorders as simple autism, enuresis, etc. (Jones, 1968; Myers, 1968), in which the child is often unable to talk about his illness.

Further research seems suggested by several aspects of this study: (a) a comparison of the effectiveness of trained versus untrained therapists in the use of systematic desensitization; (b) additional modification of the technique as a function of the nature of the symptoms treated, for example, possible treatment of paranoid schizophrenia by exposure to interpersonal interactions inducing suspiciousness; (c) determination of what acts as a reward in the therapeutic setting; and (d) the assessment of S awareness and its necessity in the successful application of systematic desensitization.

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(Received February 28, 1969)